Lecture 2 – JavaScript Data Types (Part 1)

Why Are Data Types Needed?

JavaScript me data types isliye important hain kyunki ye decide karte hain ki **kisi variable me kis type ka data store hoga** aur uske sath kaunse operations possible hain.

Example:

- Instagram comment → ***string***
- Like count → ***number***
- Block status → ***boolean***

Example Code :

```
let comment = "Nice pic!";
console.log(comment);          // Output: Nice pic!
console.log(typeof comment);   // Output: string
```

Types of Data Types in JavaScript

1. **Primitive Data Types –** *Simple & immutable*
2. **Non-Primitive Data Types –** *Complex like Arrays, Objects, Functions*

👉 Is lecture me sirf **Primitive Data Types** cover kiye gaye hain.

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1. Number

- Represents **integers and decimals**
- **Use Case:** Age, balance, likes

Example :

```
let balance = 5000.75;
console.log(balance);           // Output: 5000.75
console.log(typeof balance);    // Output: number
```

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2. String

- Represents **textual data** inside `' '`, `" "`, or `` ` ` ``
- **Use Case:** Username, comments, messages

Example :

```
let username = "Harshal";
console.log(username);          // Output: Harshal
console.log(typeof username);   // Output: string

let post = "Harshal is a good  boy";
console.log(post);              // Output: Harshal is a good boy
console.log(typeof post);       // Output: string
```

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3. Boolean :

- Represents **true/false**
- **Use Case:** Login status, block status, admin check

Example :

```
let isBlocked = true;
console.log(isBlocked);          // Output: true
console.log(typeof isBlocked);   // Output: boolean
```

4. Undefined

- Variable is **declared but not assigned**
- **Use Case:** Input field empty, data not received

Example :

```
let searchInput;
console.log(searchInput);        // Output: undefined
console.log(typeof searchInput); // Output: undefined
```

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5. Null

- **Intentionally empty** value
- **Use Case:** Server down response, unavailable data
  Jab **Server Down Hoga** Tab **0** (Zero) Ko Reply me Dega Nahi Isliye Waha Pe **Null** Send Kar Dega

Example :

```
let result = null;
console.log(result);            // Output: null
console.log(typeof result);     // Output: object
```


- **Intentionally empty** value
- **Use Case:** Server down response, unavailable data
  Jab **Server Down Hoga** Tab **0** (Zero) Ko Reply me Dega Nahi Isliye Waha Pe **Null** Send Kar Dega

⚠️ **Note:**
`typeof null` returns `"object"` → **JavaScript bug** preserved for backward compatibility

🧠 `typeof null` → "object" क्यों होता है?

- **JavaScript का एक पुराना bug** है।
- पहले type **memory address देखकर detect** होता था — और `null` भी **object-type memory pattern** follow करता था।
- अगर इसे अब **ठीक किया जाए तो existing पुराना code टूट सकता है**।
- इसलिए **backward compatibility** के लिए इसे वैसे ही रखा गया है।

🟩 **Conclusion:**
`typeof null` **"object" दिखाता है — ये एक पुरानी गलती है जिसे compatibility की वजह से अब भी maintain किया जाता है।**

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6. BigInt :

```
let bigValue = 123456789123456789123456789n;
console.log(bigValue);          // Output: 123456789123456789123456789n
console.log(typeof bigValue);   // Output: bigint
```

- Used for **very large integers**
- **Use Case:** Aadhaar number, cryptographic keys

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7. Symbol

- Represents a **unique and immutable identifier**
- **Use Case:** Object keys, hidden/private values

Example :

```
let userId = Symbol("id");
console.log(userId);            // Output: Symbol(id)
console.log(typeof userId);     // Output: symbol
```

Summary Table

Data Type	Definition	Example	typeof
Number	Integer or decimal values	`let x = 45.6`	`"number"`
String	Text in quotes	`let name = "Raj"`	`"string"`
Boolean	True/false value	`let isLive = false`	`"boolean"`
Undefined	Declared but not assigned	`let y;`	`"undefined"`
Null	Intentional absence of value	`let z = null;`	`"object"`
BigInt	Large integer values (suffix `n`)	`let big = 123n`	`"bigint"`
Symbol	Unique identifier	`let key = Symbol("uid")`	`"symbol"`